Dynamics of a Bio-Inspired Dual-Auger Burrowing Robot in Granular Media: DEM-MBD Cosimulation with Complementary Kinematics-Controlled Tests

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Abstract: This study investigates the dynamics of a bioinspired dual-auger burrowing robot in granular media using a discrete element method-multibody dynamics (DEM-MBD) cosimulation approach, complemented by kinematics-controlled tests performed with pure DEM (MBD disabled) and DEM-MBD simulations. Observations indicate that when the same-handed augers rotate in the same direction, the robot primarily moves horizontally, accompanied by minor pitching and yawing motions. The stator plays a crucial role in providing antitorque by rolling in the opposite direction of the augers. Kinematics-controlled tests revealed key insights into the thrust, drag, and lift forces governing the robot's burrowing motion, highlighting that thrust forces redistribute once the robot starts moving. Additionally, auger rotation appears to reduce drag forces, facilitating burrowing. The observed upward and pitching movements result from the uneven distribution of lift forces induced by the augers' dragging and rotating actions. While pure DEM is suitable for kinematics-controlled scenarios, the DEM-MBD cosimulation provides a realistic, fully coupled description of burrowing mechanics by resolving both granular interactions and the robot's rigid-body dynamics. **DOI: 10.1061/JGGEFK.GTENG-13467.** *© 2026 American Society of Civil Engineers.*

Author keywords: Discrete element methods; Multibody dynamics; Bio-inspired; Robotics; Burrowing; Thrust; Drag; Lift; Pitch.

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Introduction

Various geotechnical applications such as subsurface exploration, site investigation, foundation, and excavation involve soil-structure interactions. Alternatively, various animals and plants in nature have adapted themselves to underground environments to survive, shelter, and locomote, also involving soil-structure interactions. Over time, mainly through evolution by natural selection, they have adapted biological strategies which have favored them with specific functions to interact with soil. Many of the principles derived from these biological strategies could be implemented in geotechnical engineering solutions and systems (Tao et al. 2019; Huang and Tao 2020a; Martinez et al. 2022; Le et al. 2025).

This paper concerns burrowing, which we define as moving through, penetrating in or excavating soil using cyclic body motions without external forces. Burrowing can be seen as a special type of soil-structure interaction, emphasizing on internally driven motion, in contrast to the geotechnical soil-structure interactions which are all externally driven. A broad diversity of organisms, such as animals and plants, utilize various strategies to burrow, bury and locomote underground. For instance, the Sandfish lizard (*Scincus scincus*) burrows into the soil by relying on body undulation (Baumgartner et al. 2008; Maladen et al. 2011a). Most annelids such as earthworms use a peristalsis strategy to burrow by expanding and extending the front tip (Huang and Tao 2020b; Ruiz et al. 2015; Pirrone et al. 2024; Naziri et al. 2024) and propagating a peristalsis wave backward (Dorgan 2015). Some mollusks such as razor clams (e.g., the Atlantic *Ensis ensis*, or the Pacific *Siliqua patula*) rely on a dual anchor or push-pull strategy by coordinating the motions of their shells and feet (Tao et al. 2020; Trueman 1967). Recently, these concepts, especially the dual-anchor strategy inspired many designs of burrowing robot or probes which were studied from a geomechanics perspective (Chen et al. 2021, 2023; Huang 2020; Huang and Tao 2018, 2020a, b; Ma et al. 2020; Martinez et al. 2020; Tao et al. 2019, 2020; Zhang et al. 2023; Zhang and Fuentes 2022; Zhong et al. 2023; Borela et al. 2021; Wang et al. 2024; He et al. 2024).

In this paper, we explore a different strategy involving rotation and helical motion, drawing inspirations from various biological burrowing or swimming (Fig. 1). *Erodium* and *Pelargonium* employ a self-burial mechanism facilitated by rotation, which is driven by humidity-induced helical coiling and uncoiling of their awns (Jung et al. 2014). The helical configuration of the coiled seed generates force and torque through coil expansion and awn-tail rotation, enabling digging (Stamp 1984). Burrowing in sand is also analogous to swimming in low-Reynolds number viscous fluid, where the media-structure interaction resistance dominates over the inertial effects. Of particular interests in this research is spiral-shaped spirillum bacteria, which depend on a morphological helical cell as a survival tool to diffuse in viscous environments (Young 2006). The rotation of the helical cell breaks the time-reversal symmetry and generates a net propulsive force needed for its translational swimming movement (Cohen and Boyle 2010; Qiu et al. 2014; DiLuzio et al. 2005). Similarly, the Sandfish lizard's undulatory body movement results in a sinusoidal trace; this motion can be conceptualized as a two-dimensional analog to three-dimensional helical rotation. All these inspirations hint at a helically driven burrowing robot.

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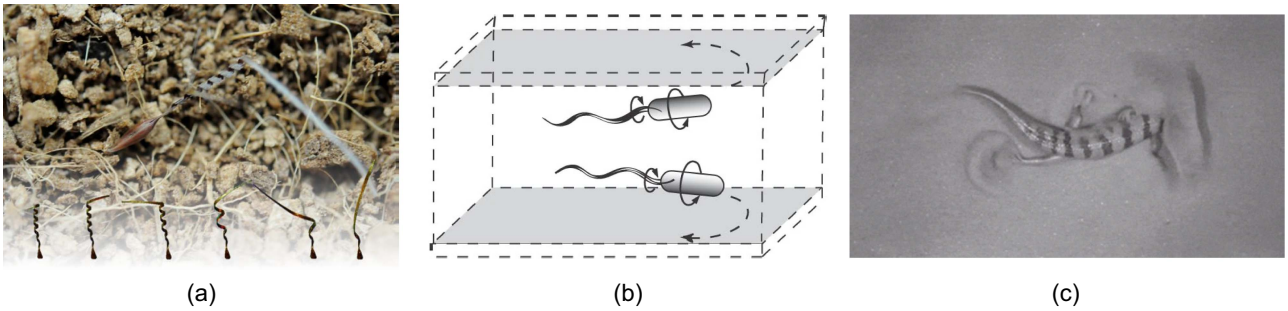

Fig. 1. Bioinspirations for the proposed burrowing robot: (a) Seed of *Pelargonium carnosum* digging into the ground due to unwinding its awn; (b) *E. coli* swims like a corkscrew; and (c) Sandfish *Scincus scincus* burrows into soil. [Reprinted (a) from W. Jung, W. Kim, and H.-Y. Kim, "Self-burial mechanics of hygroscopically responsive awns available for purchase," *Integrative and Comparative Biology*, 2014, 54 (6), pp. 1034–1042, by permission of the Society for Integrative and Comparative Biology; reprinted (b) with permission from *Springer Nature: Springer*, "Escherichia coli swim on the right-hand side," W. R. DiLuzio, L. Turner, M. Mayer, P. Garstecki, D. B. Weibel, H. C. Berg, and G. M. Whitesides, © 2005; reprinted (c) with permission of the Royal Society, from "Mechanical models of sandfish locomotion reveal principles of high performance subsurface sand-swimming," R. D. Maladen, Y. Ding, P. B. Umbanhowar, A. Kamor, and D. I. Goldman, *Journal of the Royal Society Interface*, 8 (62), 2011; permission conveyed through Copyright Clearance Center, Inc.]

Recently, we reported an experimental study on a helically driven downward burrowing robot (Bagheri et al. 2024). Yet, significant knowledge gaps exist concerning the interaction between a helically driven burrowing robot and granular media. A central question is, how the kinematics of the robot correlate to its kinetics and the granular media responses? To answer this question, this paper builds on our earlier efforts in applying coupled discrete element method (DEM)-multibody dynamics (MBD) simulations in simulating robotic movements in granular materials (Zhong and Tao 2023; Shahhosseini et al. 2023; Shahhosseini and Tao 2025). Here, we presents a more comprehensive investigation into the behavior of a dual auger self-burrowing robot, highlighting the strengths of the DEM-MBD method in capturing the robot's trajectory, the forces and torques on its components due to internal actuations, as well as granular displacements and forces. Additionally, we explore whether such complex interactions can be tackled by simpler, kinematic, DEM approach alone.

The paper starts with a description of the robotic design, an introduction of the numerical methods, and a summary of simulated cases, followed by the presentation and explanation of the observed dynamic behavior of the burrowing robot based on DEM-MBD cosimulations. We will then evaluate the benefits and limitations of the adopted simulation platform in DEM-MBD cosimulations, comparing to DEM only simulations. We conclude the paper with a brief conclusion and outlook for future research.

Methodology

Burrowing Robot Design

The most critical components of our robot are a pair of helical structures which we refer to as augers. The augers mimic the spiral tail of spirillum bacteria or a three dimensional sinusoidal trace of Sandfish lizards. The rotation of the augers is expected to provide thrust needed for burrowing. In addition to the augers, a burrowing robot should also include a stator, which provides counter torque to the helical structure. For example, in self-burying *Pelargonium* seeds and swimming spirillum bacteria, the awn bristles and the head act as the stators, respectively. We used a design consisting of four blades distributed along a hollow cylinder. To rotate the augers, two gear motors can be embedded in the stator and connected to the augers.

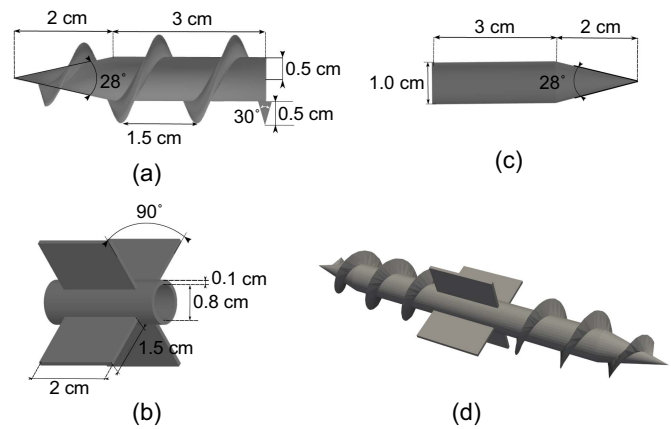

Fig. 2. Detailed dimensions of robotic parts: (a) auger, (b) stator; (c) control penetrator; and (d) assembled robot.

Figs. 2(a and b) illustrate the detailed dimensions of the augers and the stator. Specifically, the two augers are constructed by wrapping a blade around a control penetrator [Fig. 1(c)], following a helical path with right-handedness and a pitch of 1.5 cm. The four stator blades are equally spaced around the stator cylinder at 90-degree intervals. The finished look of the assembled robot with mesh surfaces is shown in Fig. 2(d).

Discrete Element Method

Discrete element method or DEM, has been widely adopted in the study of soil–structure interaction problems, such as pile installation (Shi et al. 2019), site investigation (Arroyo et al. 2011; Khosravi et al. 2020), and general interface behaviors (Jing et al. 2018). More recently, DEM has been adopted to model biological or bio-inspired penetration and burrowing (Maladen et al. 2011a; Fakih et al. 2019; Patino-Ramirez and O'Sullivan 2024; Sandeep and Evans 2023; Huang and Tao 2018, 2020a, b; Tang and Tao 2022a, b; Chen et al. 2021, 2023; Zhang et al. 2023; Zhang and Fuentes 2022; Zhong et al. 2023; Borela et al. 2021; Ma et al. 2020). For the sake of brevity and to avoid redundancy, we provide only a concise overview of the established DEM method, referring readers to existing literature for comprehensive details.

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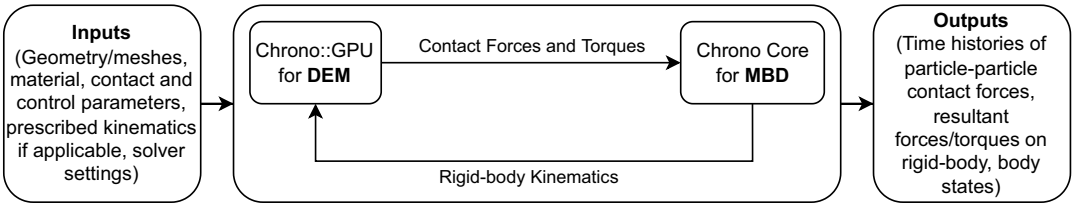

Fig. 3. DEM-MBD cosimulation workflow in Chrono.

In the DEM simulations, particles are allowed to undergo small deformations, and the primary assumption is that the mechanical behavior of the system is driven by Newtonian dynamic updates between individual entities (Campello 2018). In each time step, the normal contact force, friction tangential force, rolling friction torque, and cohesion forces are calculated based on the stiffness, damping coefficients, and contact constitutive models including the Hertzian contact theory, the Mindlin friction law, and rolling resistance models (Fang et al. 2021). The displacements of the particles are then calculated based on the calculated forces, allowing for the progression to the next time step with the updated positions of the particles. More detailed DEM formulation is included in the Supplemental Materials.

Multibody Dynamics

Multibody dynamics is the study of dynamic behavior between a finite set of elements based on mechanical principles (Schiehlen 1997). A multibody system consists of several rigid or flexible bodies and particles interconnected with joints, supports, dampers, active force, and position actuators. Each body is described by at least one body-fixed frame, and the interaction between bodies and frames is defined and constrained by kinematic joints. The joints constrain each body's degree of freedom based on the user-defined kinematics. As a result of Lagrange's equations, derived from D'Alembert's principle which is an alternative to Newton's second law, any virtual displacement consistent with the constraints of the system is equal to the sum of the difference between the forces acting on the system and the inertial force of the system. (Schiehlen 1997). A detailed mathematical description of the MBD method can be found in the Appendix.

DEM-MBD Cosimulation: Open-Source Project Chrono

While DEM can be applied to study the behavior of mechanical systems in a granular media (Thoesen et al. 2019; Huang and Tao 2020b), it results in a kinematics-controlled simulation in which the motion of the robotic object does not necessarily obey laws of motion. Instead, the robot's movement is based on the prescribed velocity and position (Shahhosseini et al. 2023). In contrast, MBD describes a multibody system by dynamically analyzing equations of motion (Mazhar et al. 2013), as described in the proceeding section. Therefore, a DEM simulation can be referred to as a kinematics-controlled system, while MBD simulations are instead dynamics-based. This study employs a coupled DEM and MBD approach, implemented in Project Chrono 7.0.0 (hereafter referred to as Chrono), to investigate the burrowing mechanism of a bio-inspired, self-burrowing robot. Chrono is an open-source physics-based platform that allows the DEM-MBD cosimulation, among other capabilities (Mazhar et al. 2013; Tasora et al. 2016). In Chrono, particles are simulated as discrete elements; the user can also import meshes to represent the geometry of a multibody system interacting with granular material (Fang et al. 2021).

Chrono consists of several simulators or modules, and the Chrono::GPU engine (the DEM module) and the Chrono core engine (including MBD module) can be coupled together for cosimulations (Fig. 3). First, the Chrono::GPU engine computes the particle-particle and particle-mesh contact forces and torques based on the Hertzian contact model. The particle-mesh contact forces and torques are then transferred to the Chrono core module as external API forces for collision detections in which the dynamics of the mesh objects are simulated to update their position and velocity information. These updates are then transferred back to the Chrono::GPU engine to update the dynamics of the granular media based on Newton's second law and API mesh managers for the next time step (Fang et al. 2021). The collision detection in Chrono takes part in two phases, narrow and broad. The broad phase studies only the possibilities of contact between two particles and not actual encountering, while the narrow phase calculates the actual occurrence of potential contacts (Mazhar et al. 2013).

Contact detection and force calculations are among the most time-consuming and expensive steps in the robotic-granular simulation due to the small time step and the large number of particles. Chrono tackles this problem by leveraging graphics processing unit (GPU) to take advantage of its global memory (Fang et al. 2021). Moreover, when the interactions are in short range, as in particle-particle contacts, there is no need to calculate the contact force between two elements far from each other. Instead, the linked cell method divides the main domain into smaller subdomains containing particles with the exact origin or in neighbor (Fang et al. 2021). In Chrono, the current size of subdomains is 3.5 times the dimension of a discrete element but can be modified by the user according to the available memory in GPU and the size of the CUDA block thread (Fang et al. 2021). In addition, Chrono::GPU splits each mesh into a collection of triangular facets interacting with particles in each subdomain. Then, the total force and torque on all triangles representing the objects are calculated using the Hertzian contact force model and then transferred back to the Chrono core engine to begin the calculation cycle. The Hertzian contact force model requires essential variables, including normal/tangential stiffness, normal/tangential damping, and static friction coefficient of particle-particle and particle-mesh, to calculate normal and tangential forces.

The DEM engine Chrono::GPU has been rigorously validated by the development team through wave propagation, rotating drum, direct shear, and crater tests; its coupling to MBD has been demonstrated with the simulation of the NASA Curiosity Rover (Fang et al. 2021). Its ability to simulate granular behavior has also been validated in the context of soil mechanics (He et al. 2021; Zhong and Tao 2023).

Model Construction

Granular Sample Preparation

Monosized spherical particles with a diameter of 3 mm was used to prepare the granular bed. The particles model glass beads that used

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in an previous experimental setup (Zhong and Tao 2023). The sample is prepared by depositing particles into a container (41.7 cm L× 25.7 cm W) with a Poisson disk sampling (PDS) method (Bridson 2007), where particles are simulated with a layer distance of 2.04 times the particle radius. The PDS method generates a randomized yet well-spaced distribution of particles. This approach prevents clustering and avoids the artificial ordering.

To ensure a desired medium-dense sample, the friction coefficients between particles and between particles and the mesh are first set to 0.08 and 0.15, respectively. Following a particle settling period of 5 s (Phase I), the friction coefficients are readjusted to 0.16 and 0.3, respectively, and left to settle for an additional 2 s (Phase II) (Thoesen et al. 2019; Shahhosseini et al. 2023).

Using the theoretical bounds for monodisperse spheres packings ($e_{\min} = 0.35$, $e_\max = 0.91$) (Mitchell and Sog 2005), the sample's relative density is approximately 56%. If, instead, the random close packing ($e_{\min} \approx 0.56$) and random loose packing ($e_{\max} \approx 0.82$) limits are adopted (Jerkins 2008; Onoda 1990; Scott and Kilgour 1969), the estimated relative density is about 87%.

A total of 602,387 particles were used to prepare a sample with a height of 13.6 cm. The sample exhibits a predominantly uniform relative density with natural spatial variations, except near the corners, where significantly lower packing densities are observed (Fig. 4). There are no visible signs of artificial ordered packing. The sample is sufficiently large to minimize boundary effects, as discussed in "Results."

Due to the inclusion of the robot and the free fall deposition method, the top surface of the sample is not even and not further treated to preserve its original state. The particle position, angular velocity, and rotational position are recorded and saved as a checkpoint file, which can be subsequently utilized to initiate each simulation case.

Robot Setup and Control

The robot parts include 1,828 triangle meshes. In Chrono, these robot part meshes are first imported and connections are made among different bodies: both augers are connected to the stator using rotational motor links, thus constraining the motion of the augers. The robot meshes are originally positioned between two sets of particle layers above the container, and are deposited together with the particles so that the robot would be embedded in the sample after deposition. The resulting embedment depth of the robot is about 6 cm. The rotation speed and direction can be specified as needed. In this study, whenever rotation is applied, a speed of 60 revolutions per minute (rpm) in the counterclockwise direction, as viewed from the rear along the robot's long axis, is used.

Simulation Cases

A total of five test cases are conducted to study the self-burrowing behavior of the robot, the effect of helical design, the drag and thrust forces (Table 1).

In the burrow test case, only the axial rotation of the augers are controlled to a constant speed of 60 rpm; the locomotion of the robot is a result of the coupling between the DEM and MBD modules. Thus, the robot movements, including the translation and rotation, are induced. In the control test case, the augers are replaced with two smooth cylinders without the helical blades, to highlight the effect of the geometry and rotation.

In the thrust test case, the augers of robot rotates at 60 rpm while it is held in place by prohibiting the stator's movements. The purpose is to understand the driving thrust force generated by the rotation of augers. In the drag case, the stator is horizontally dragged with a prescribed velocity of 1.4 cm/s while the axial rotation of the augers and the tilting (or rotation around the x and z axes) of the robot are prohibited. The purpose is to understand the resistive drag forces on the robot with nonrotating augers. The rotdrag test case is the same as the drag case, except that the augers are rotated counterclockwise at 60 rpm. The purpose is to understand how rotation affects drag forces.

To better understand the suitability and limitations of DEM-MBD cosimulation, both DEM and DEM-MBD approaches were utilized in the kinematics-controlled drag, rotdrag, and thrust test cases. The pure DEM method was achieved by deactivating MBD

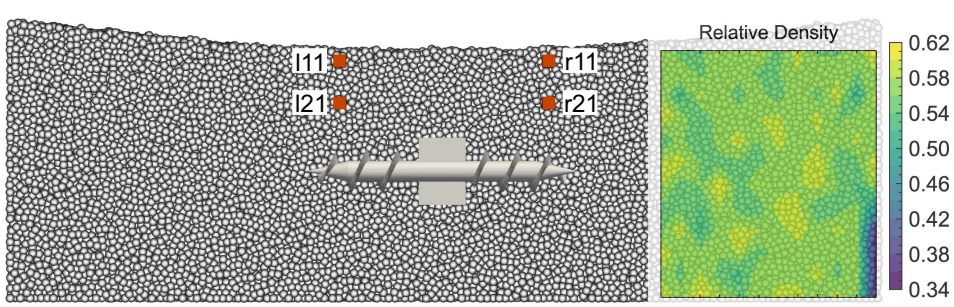

Fig. 4. Cross-sectional view of the simulation setup following sample preparation, featuring regions of interest (l11, l21, r11, r21) for particle displacement analysis, highlighted by boxes; and overlay of the relative density distribution, spatially aligned with the sample geometry. Note that the relative density was calculated based on theoretical minimum and maximum void ratios for monosized spherical particle packings.

Table 1. Summary of simulation cases

Test cases	Axial rotation	Translation	Tilt	Methods[a]	Purpose
Burrow	Controlled	Induced	Induced	1	Self-burrowing behavior
Control	Controlled	Induced	Induced	1	Effect of geometry
Drag	Prohibited	Controlled	Prohibited	1, 2	Drag force with nonrotating augers
Rotdrag	Controlled	Controlled	Prohibited	1, 2	Drag force with rotating augers
Thrust	Controlled	Prohibited	Prohibited	1, 2	Thrust force from rotating augers

[a]1 = coupled DEM-MBD approach; and 2 = pure DEM approach.

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functionality in Chrono, where the robotic parts' motion was defined through kinematics-controlled motions. The transitional and rotational velocities of the augers and stator were prescribed using built-in functions in Chrono. In these simulations, the robotic parts could affect the granular media, but the granular media could not induce the robot's motion. In contrast, under the coupled DEM–MBD cosimulation, the robot's motion in these kinematics-controlled cases was computed dynamically within the fully coupled framework, while concurrently being constrained by the prescribed kinematics. Unfortunately, this treatment introduced internal conflicts within the solvers, leading to variability in the calculations, which we will examine later.

Each simulation case has been run twice, totaling 16 simulations overall. For each 5-s simulation, a time step of 10^{-6} second is used. With an NVIDIA V100 GPU, it takes 14–16 h for the cosimulations and 5–8 h for the pure DEM simulations. The material and interaction properties of the simulated particles and meshes is provided in the Appendix. The granular media properties represent that of glass beads, and the mesh properties represent that of 3D-printing surfaces using acrylonitrile butadiene styrene. These properties were based on experimental data in the literature (Thoesen et al. 2020), which were verified in Chrono from previous numerical studies (Zhong and Tao 2023; Shahhosseini et al. 2023).

Data Collection and Analysis

With the built-in helper functions from Chrono, we collected the data on the evolving position $[P_x, P_y, P_z]$ of the centroids of all the robot parts as well as the resultant force $\mathbf{F}$ ($[F_x, F_y, F_z]$) and torque $\mathbf{T}$ ($[T_x, T_y, T_z]$) on each robot part as regard to their centers from the contacting particles. We also collected the positions of the particles as well as contact forces between any pair of contacting particles. Hereafter, the left auger, stator, and right auger are denoted as L, S, and R, respectively. Based on these data, we derived the kinematics and kinetics of the robot (Fig. 5).

Kinematics Data

The displacement of the robot, $[D_x, D_y, D_z]$, can be expressed as the difference between the stator's centroid position at a given time, $[P_{xS}, P_{yS}, P_{zS}]$ and its original position, $[P_{xS0}, P_{yS0}, P_{zS0}]$ [Eq. (1)]

$$[D_x, D_y, D_z] = [P_{xS}, P_{yS}, P_{zS}] - [P_{xS0}, P_{yS0}, P_{zS0}] \tag{1}$$

The pitch and yaw of the robot can be calculated using the coordinates of centroids of any two robotic parts. The centroids of left and right augers, C_L and C_R, are used to calculate the pitch and yaw of the robot.

As shown in Fig. 5(b), within a Cartesian coordinate system (x, y, z), the robot's long axis, denoted as $\mathbf{C_L C_R}$, was initially aligned with the y axis. At any subsequent moment, if $\mathbf{C_L C_R}$ deviates from the y axis, pitch and/or yaw is present. Pitch can be defined as the angle between $\mathbf{C_L C_R}$ and its projection, $\mathbf{C_L C'_R}$, on the horizontal plane (xy). Yaw is defined as the angle between $\mathbf{C_L C'_R}$ and the y axis.

Using trigonometry and the coordinates of each centroid, the pitch and yaw angles can be calculated as follows:

$$\text{pitch} = \arctan\left(\frac{|\mathbf{C_L C_R} \times \mathbf{k}|}{\mathbf{C_L C_R} \cdot \mathbf{k}}\right) = \arctan\left(\frac{P_{zR} - P_{zL}}{\sqrt{(P_{xR} - P_{xL})^2 + (P_{yR} - P_{yL})^2}}\right) \tag{2}$$

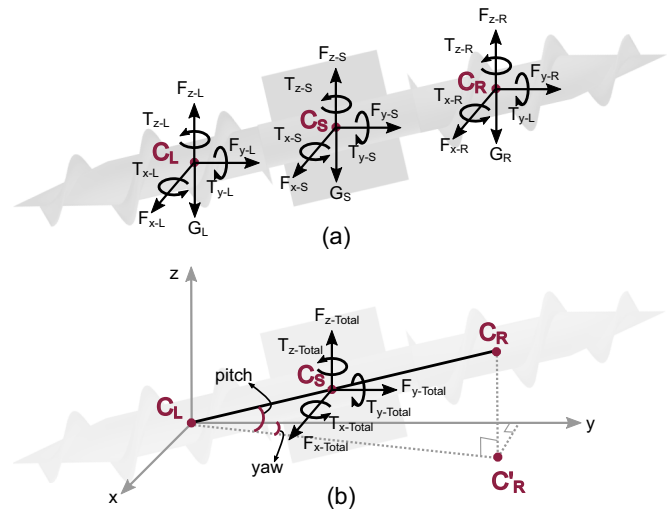

Fig. 5. Kinetics and orientation of the robot at the end of burrowing. CL, CS, and CR denote the centroid of the left auger, right auger, and stator, respectively: (a) forces and torques on robotic parts; and (b) the total force and torque on the entire robot, and the illustration of the robot orientations (pitch and yaw). Note that the orientation in the figure has been much exaggerated to show the angles clearly.

$$\text{yaw} = \arctan\left(\frac{\mathbf{C_L C_R} \cdot \mathbf{i}}{\mathbf{C_L C_R} \cdot \mathbf{j}}\right) = \arctan\left(\frac{P_{xR} - P_{xL}}{P_{yR} - P_{yL}}\right) \tag{3}$$

In addition to pitch and yaw, the stator also roll around the long axis of the robot. To calculate the roll angle, an additional point which is not collinear with the long axis is needed. Here, a vector ($\mathbf{V}$), which lies on one of the blade surfaces and is also perpendicular to the long axis of the robot, is tracked. Using its initial and present positions, $\mathbf{V}_0$ and $\mathbf{V}_t$, respectively, the roll angle can be calculated

$$\text{roll} = \arccos\left(\frac{\mathbf{V}_0 \cdot \mathbf{V}_t}{|\mathbf{V}_0||\mathbf{V}_t|}\right) \tag{4}$$

Kinetics Data

Considering the gravity of the robotic parts ($\mathbf{G}_L, \mathbf{G}_S, \mathbf{G}_R$), the total force $\mathbf{F}_{total}$ ($[F_{x-total}, F_{y-total}, F_{z-total}]$) on the robot can be expressed as one acting on its overall centroid or the center of the stator [Eq. (5)]

$$\mathbf{F}_{total} = \mathbf{F}_L + \mathbf{F}_S + \mathbf{F}_R + \mathbf{G}_L + \mathbf{G}_S + \mathbf{G}_R \tag{5}$$

Considering the position vectors from the stator centroid C_S to the augers centroids C_L and C_R, denoted as $\mathbf{C_S C_L}$ and $\mathbf{C_S C_R}$, respectively, the total torque $\mathbf{T}_{total}$ ($[T_{x-total}, T_{y-total}, T_{z-total}]$) can be expressed using Eq. (6)

$$\mathbf{T}_{total} = \mathbf{T}_L + \mathbf{T}_S + \mathbf{T}_R + \mathbf{C_S C_L} \times \mathbf{G}_L + \mathbf{C_S C_R} \times \mathbf{G}_R \tag{6}$$

Results

Burrowing Case: Dynamics of the Dual-Auger Burrowing Robot

Kinematics: Robot Motion

As the augers began rotating counterclockwise around the positive y-axis at 60 rpm, the robot gradually initiated both translational and rotational movement, as demonstrated in Video 1 in the

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Supplemental Materials. Fig. 6 illustrates the displacement of the robotic components—stator (S), right auger (R), and left auger (L)—across three axes. During the burrowing process, the robot gradually moved forward, slightly drifted to its right, and ascended and tilts toward the surface.

Over a 5-s span, the y-axis displacement, reflecting forward translational motion, shows a total advancement of 5.2 cm for all robotic parts and the velocity is more or less constant [Fig. 6(b)].

The movement along the x-axis, reflecting its sideways motion, is relatively small, with only a 0.08 cm displacement for the stator [Fig. 6(a)]. The differential displacements between the right and left auger in the x-axis direction indicates rotation around the z-axis, or yawing. The calculated yaw angle [Eq. (3)] gradually increases to about −1.5 degrees, with negative angle meaning counterclockwise around the positive z direction Fig. 6(d).

The z-axis displacements shows that the stator ascends by 0.15 cm, the right auger by 0.36 cm, and the left auger descends by 0.08 cm [Fig. 6(c)]. The differential z-axis displacement of 0.44 cm between the right and left augers indicates a pitching motion of the robot. The pitch angle [Eq. (2)] almost linearly increases to a maximum of about 4 degrees at the end of the simulation [Fig. 6(d)].

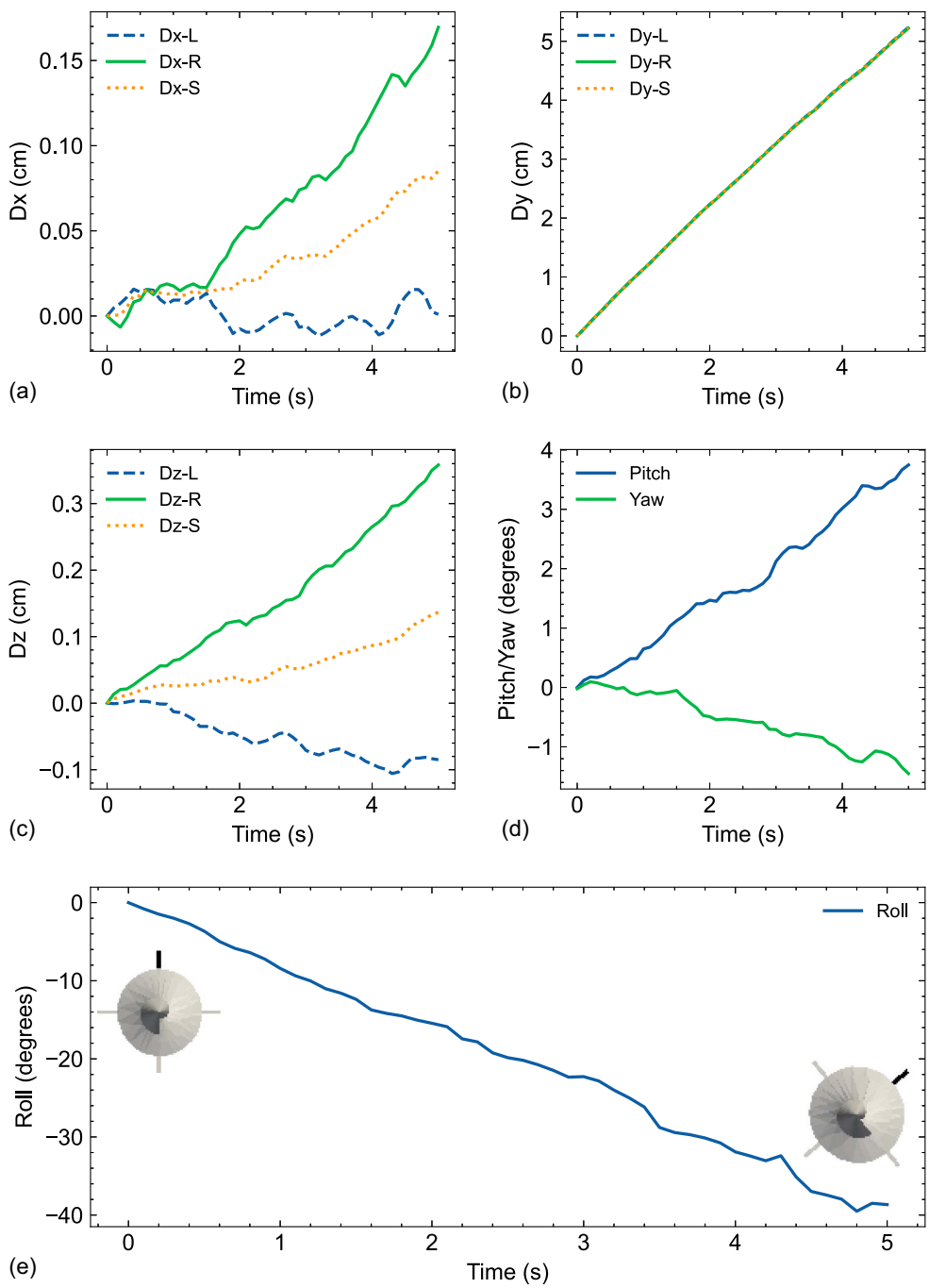

Fig. 6. Burrowing kinematics: (a) Dx: movement in the transverse direction; (b) Dy: movement in the longitudinal direction; (c) Dz: movement in the vertical direction; (d) pitch: rotation around the x axis; yaw: rotation around the z axis; and (e) roll: rotation around the robot long axis; the inserts are snapshots taken at time 0 and 5 s, viewing from the rear along the robot's long axis.

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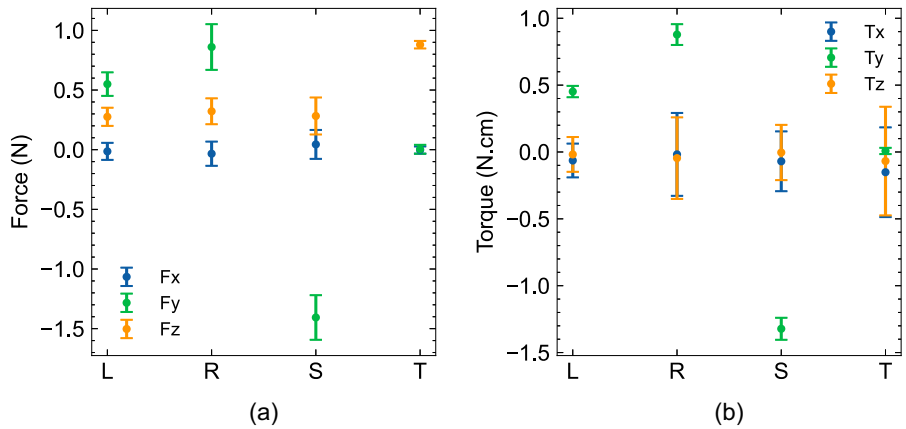

Fig. 7. Statistics of the DEM-MBD cosimulation results for one burrow case. The dots represent the average values in the steady state stage (1–5 s) and the bars represent the corresponding standard deviation. L, R, S, and T denote the left auger, right auger, stator, and total.

In addition to yawing and pitching, the stator also experienced rolling motion. While both augers rotated counterclockwise when viewed from the rear along the robot's long axis, the stator rolled clockwise, generating counter-torque to balance the system. The roll angle [Eq. (4)] increases linearly to a maximum of about 40 degrees at the end of the simulation [Fig. 6(e)]. Ideally, a stator would remain stationary. However, mobilizing resistance in the soil requires some displacement, resulting in the stator's gradual accumulation of rotation over time.

Kinetics: Forces and Torques

The kinematics of the robot resulted from the kinetics, or the forces and torques acting on the robot. A summary of the average and standard deviations of all the forces in one burrow trial is presented in Fig. 7.

Forces

Forces are responsible for the translational movement of the robot. As shown in Fig. 8, all force components fluctuate because of the discrete nature of the granular media and intermittent particle contact, acceleration, and discharge along the helical blades. Both the magnitude and the variability of F_y are the largest, as also indicated by the mean and standard deviation in Fig. 7(a). At the particle scale, the rotating blades repeatedly form and break particle–mesh contacts, generating short-lived force chains that carry load and then collapse. Owing to the helical geometry, lateral ($\pm x$) and vertical ($\pm z$) components tend to cancel by symmetry, whereas the axial component (F_y), including its instantaneous fluctuations, adds rather than cancels, producing axial thrust pulses.

The F_x values on the three robot parts are small and all fluctuate around zero, resulting in a total force that is also nearly zero.

The F_y values on the two augers are positive, showing that they are the driving forces for the forward motion of the robot. The F_y on the right auger is higher than that on the left. In contrast, the negative F_y force on the stator suggests that it experiences resistance from the particles as the robot moves forward. The total F_y forces on the robot is nearly zero, meaning the driving forces from the augers are balanced by the resistance. This balance is consistent with the nearly linear trend in the robot's y-direction displacement [Fig. 6(b)].

The F_z values on the three robot parts are predominantly positive, reflecting the lifting effect of the particles due to the robot's motion. The sum of these forces is almost balanced by the gravity [Fig. 8(d)].

Torques

The torques T_x, T_y and T_z are responsible for the robot's pitching, rolling and yawing, respectively. All torques exhibit fluctuations (Fig. 9) and display an approximately periodical pattern with a frequency of about 1 HZ, consistent with the auger rotation speed of 60 rpm.

The total torques related to pitching (T_x) and yawing (T_z) are smaller but exhibit greater fluctuations compared with T_y. The positive T_y on the augers is counterbalanced by the negative torque on the stator, confirming the antitorque function of the stator, which aligns with its rolling behavior [Fig. 6(e)]. The magnitude of the torque T_y on the front auger is consistently higher than that on the back auger.

A consistent observation is that the total force and torque on the robot is near zero. This supports the common assumption that inertial effects are negligible when studying locomotion in granular media (Darbois Texier et al. 2017; Hosoi and Goldman 2015).

Control Case: No Thrust But Lift

We conducted a control burrowing test using smooth rods (no blades) while holding all other parameters identical to the augered cases. The observed kinematics and measured forces isolate the contribution of the helical geometry to thrust production and lift generation.

As shown in Fig. 10(a), axial rotation of two smooth rods produces no translational motion in the burrowing test. Consistently, no thrust force is measured on the rotating rods [Fig. 10(b)]. Thus, the augers' helical geometry is required to generate thrust and enable effective horizontal propulsion through soil. However, a net lift force is observed in the control test, indicating that axial rotation itself, rather than helical geometry, generates lift. Further consideration of this matter is presented in the "Discussion" section.

Kinematics-Controlled Cases: Drag, Thrust and Rotdrag Tests

In all other cases, one or more degrees of freedom (DoFs) of the robot are constrained and the kinematics of the robot is prescribed. These cases are studied to better understand the causes of the burrowing behavior observed in the burrowing scenario. Both DEM-MBD cosimulations and pure DEM simulations are used to model these cases. A finding from the comparisons is that limiting one or more DoFs can sometimes lead to unstable results in the cosimulations.

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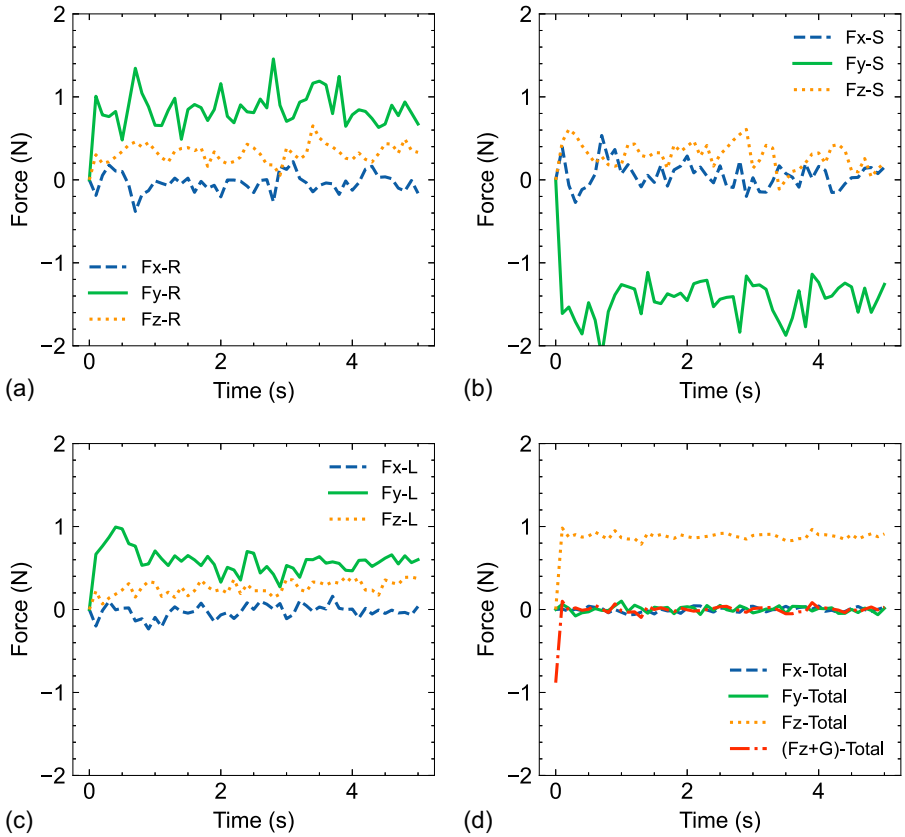

Fig. 8. Resultant particle forces on the robot parts: (a) right auger; (b) stator; (c) left auger; and (d) whole body. (Fz + G) represents the sum of the vertical mesh-particle contact forces and the gravity.

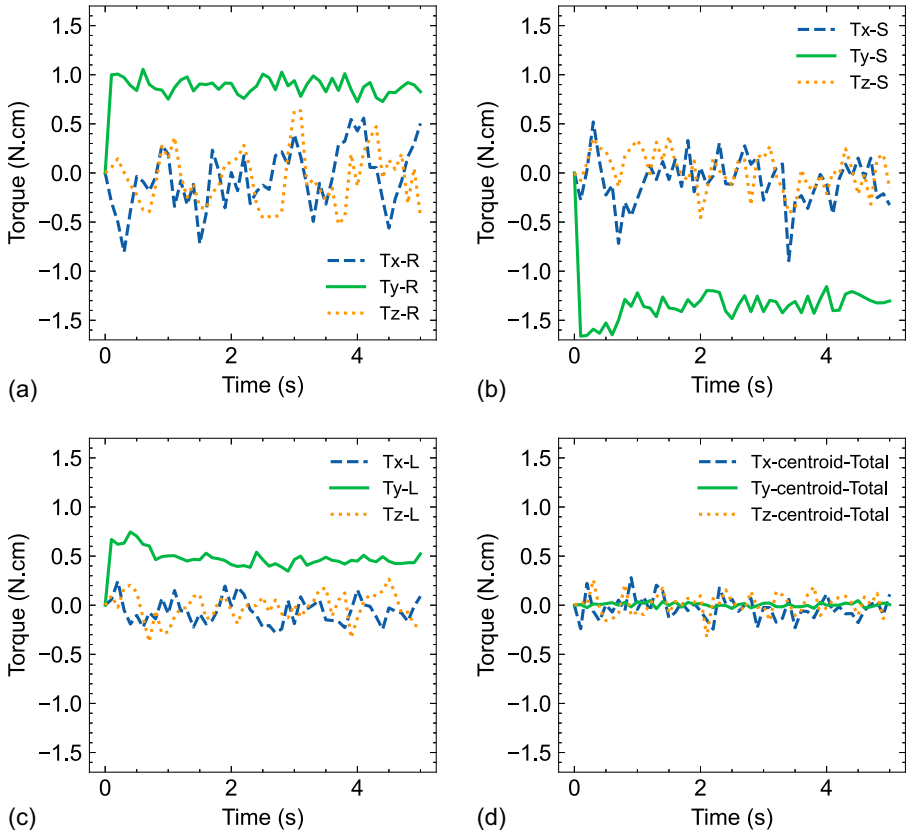

Fig. 9. Resultant torque on the robot parts: (a) right auger; (b) stator; (c) left auger; and (d) whole body.

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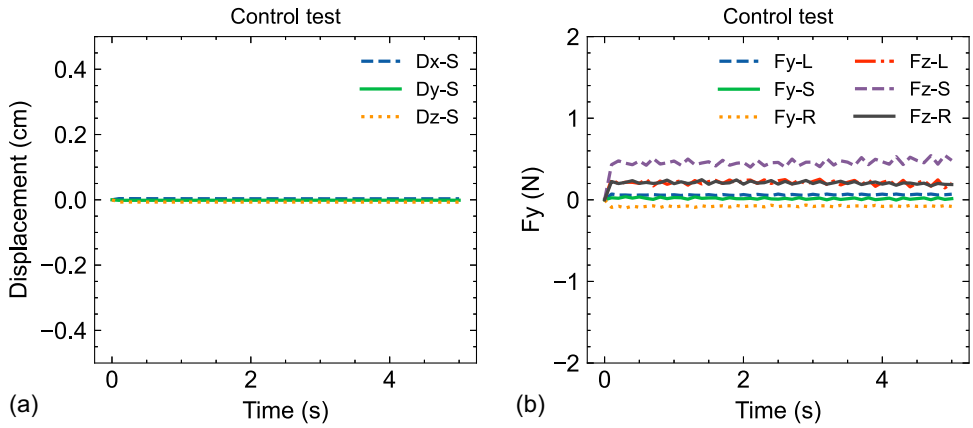

Fig. 10. Control burrowing test without auger blades: (a) displacements of robot parts; and (b) forces on the robot parts. Only the y and z components are shown for simplicity.

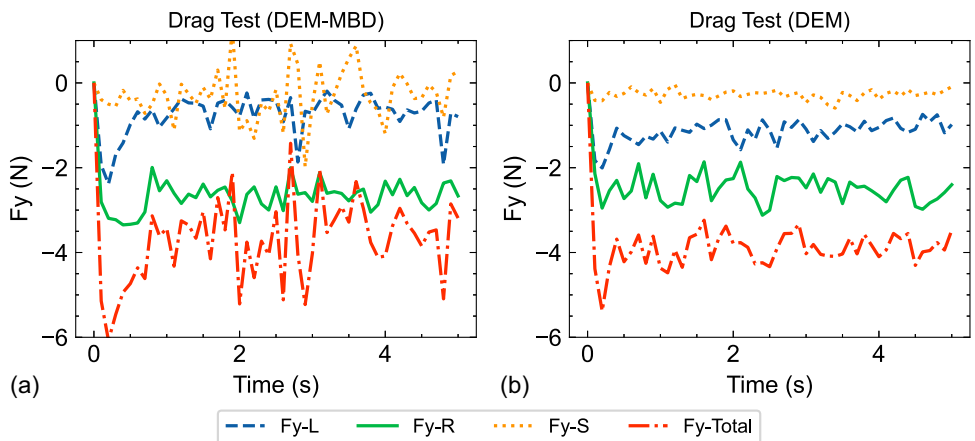

Fig. 11. Drag test results: (a) DEM-MBD coupled simulations; and (b) DEM simulations.

Therefore, we begin this section by comparing the results from the two approaches. We then focus on the results on thrust force F_y and lifting force F_z, since they are the driving forces for the significant horizontal and vertical movements.

Comparison between DEM and DEM-MBD Results

When examining the force data in the y-direction from the drag tests (Fig. 11), the overall trend and magnitude of the forces on each robot part are similar between the two approaches. However, the data from the pure DEM approach is significantly more stable, particularly for the stator part and the drag/rotdrag cases, compared to the DEM-MBD approach.

Two trials are conducted for each simulation. The mean and standard deviation of the 1–5 s data are calculated for each case and plotted according to the test cases and approaches used (see Fig. 12).

The results indicate greater run-to-run variability in the kinematics-controlled tests when using the coupled DEM–MBD approach, whereas the pure DEM approach yielded more consistent and stable outcomes. The source of this increased variability is not definitively established. However, consultation with the code developers suggests that, in kinematics-controlled DEM–MBD simulations, default motor and link settings can interact with the imposed constraints, introducing numerical conflicts and potential instabilities. Mechanistically, at each time step the DEM solver computes particle–mesh contact forces from the prescribed displacement; when these forces are passed to the MBD module, it attempts to update the rigid-body state even though the pose is constrained by the imposed trajectory, prompting repeated corrective adjustments and amplifying variability. The variability reported here arises only in kinematics-controlled DEM–MBD runs and does not reflect a deficiency of the validated cosimulation framework for fully coupled, free-motion dynamics. Accordingly, for subsequent analyses of kinematics-controlled benchmarks, we report the more stable pure-DEM results (with MBD disabled). Video 2 in the Supplemental Materials demonstrate particle velocity and robot movement during the kinematics-controlled tests (thrust, drag, and rotdrag) using the pure DEM approach, as well as the burrow test using the DEM-MBD approach.

Thrust and Drag Forces

By examining the F_y component across all cases (see Figs. 12 and 13), we can gain insights into the thrust and resistance experienced by the burrowing robot.

In the Thrust test, the left auger generates a higher thrust force (approximately 1.99 N) compared to the right auger (approximately 1.44 N). Conversely, in the drag test, the right auger experiences greater drag forces (approximately −2.5 N on the right versus −1.1 N on the left), consistent with findings from typical penetration tests. Based on these observations, one would expect the right auger to experience lower overall contact forces (arithmetic sum

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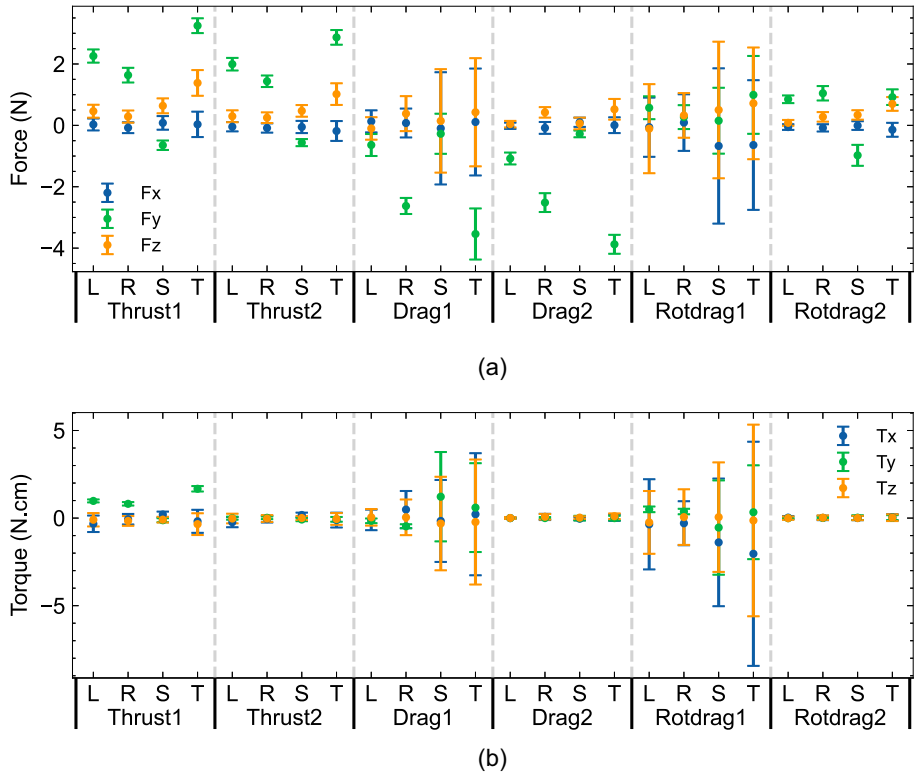

Fig. 12. Comparison of results from DEM-MBD coupled simulations (denoted as 1) and DEM simulations (denoted as 2). The dots represent the average values in the steady state stage (1–5 s) and the bars represent the corresponding standard deviations. L, R, S, and T denote the left auger, right auger, stator, and total.

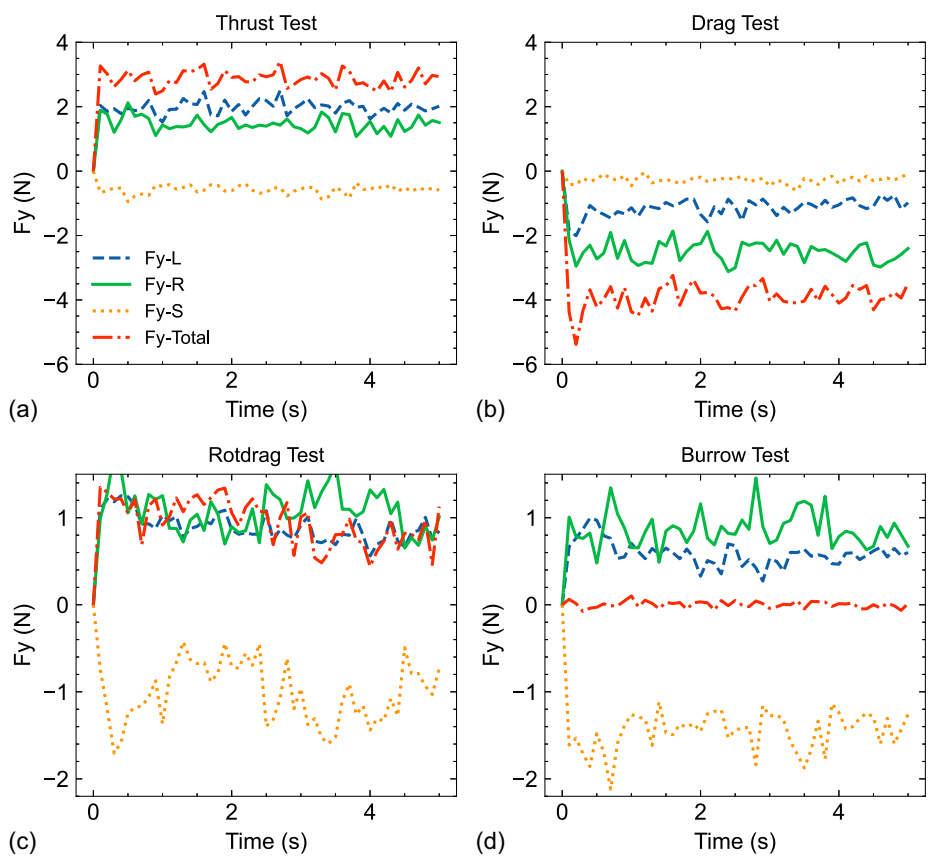

Fig. 13. y-direction forces comparison across cases: (a) thrust test; (b) drag test; (c) rotdrag test; and (d) burrow test.

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considering the signs) from the particles in the rotdrag test or burrow case. However, the data reveals the opposite: in both cases, the right auger experiences higher positive forces from the particles, with approximately 1.05 N on the right versus 0.85 N on the left in the rotdrag test, and approximately 0.86 N on the right versus 0.55 N on the left in the burrow case. This positive force can be interpreted as the net thrust force, representing the difference between the actual thrust (the force contributing to propulsion) and drag (the force contributing to resistance) experienced by the augers.

To get a better understanding, it is necessary to separate the net force on the augers into true thrust and drag forces, which is challenging, if not impossible. Insights might be gained by examining the individual particle-mesh contact forces, such as identifying which contact forces resist or promote forward motion. Unfortunately, such detailed information is not available from Chrono 7.0.0, which only records the total contact forces on the mesh bodies.

Nevertheless, we provide conceptual explanations for these observations by considering particle movements, particle contact force networks, and evidence from related research (see the "Discussion" section).

Lift Forces

By examining the F_z component across all cases (see Figs. 12 and 14), we can gain insights into the upward and pitching movements of the burrowing robot.

During the drag test, upward forces were observed on all parts of the robot: approximately 0.4 N on the right auger, 0.06 N on the stator, and 0.04 N on the left auger. This observation aligns with the drag-induced lift seen when dragging rods or objects with various geometries through granular media (Ding et al. 2011; Maladen et al. 2011b). This lift is primarily due to the asymmetric distribution of stresses on the front surface of the object being dragged.

In the thrust test, significant upward forces were observed on all robot parts: approximately 0.25 N on the right auger, 0.47 N on the stator, and 0.3 N on the left auger. A comparable lift force is observed in the control burrowing test with smooth rods and no blades [Fig. 10(b)], underscoring that axial rotation alone can generate lift.

In conclusion, both the translational and rotational motions of the augers generate lift forces that contribute to the robot's upward movement. Additionally, the uneven distribution of lift forces, particularly in the case of drag-induced lift, is responsible for the robot's pitching motion.

Particle Displacement and Contact Forces

Although we do not have access to individual particle-mesh contact forces, we can examine the particle displacements and particle-particle contact forces to gain more insights.

Temporal displacements were analyzed by examining the homogenized particle displacements within regions of interest (ROIs), which are cubic with widths of 6 mm (see Fig. 4). Given the particle diameter of 3 mm, each region can contain approximately six particles at any given time.

Fig. 15 presents the vertical homogenized particle displacements within four selected ROIs across all tests. See the Supplemental Materials for additional directions and ROIs.

Accordingly, Fig. 16 displays the accumulated displacement vectors of particles within a vertical slice of the sample. This slice lies in the lengthwise yz-plane and has a thickness of 2.5 cm along the sample width direction (x-axis). It spans the thickness of the estimated influence zone (see Supplemental Materials for details). The particle-particle contact force network at the last time step (5 s) for each case are shown in Fig. 17.

In the Thrust test, particles in the upper right regions (r11, r21) of the robot move downward [see Fig. 15(a)]. In contrast, particles

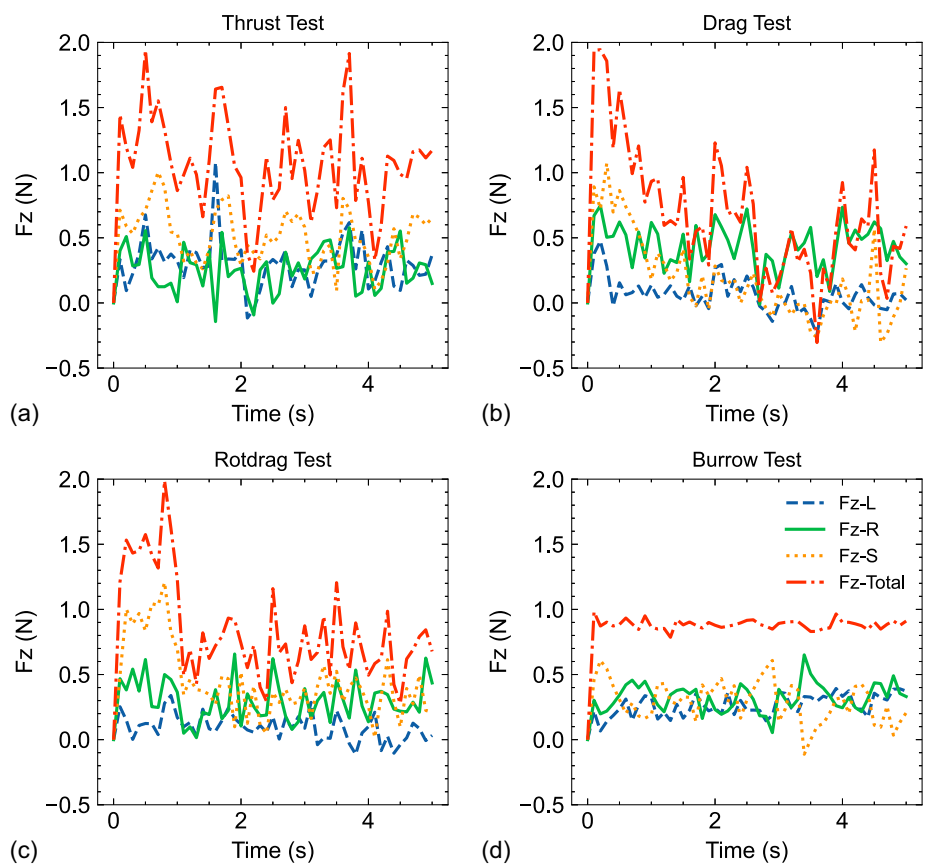

Fig. 14. z-direction forces comparison across cases: (a) thrust test; (b) drag test; (c) rotdrag test; and (d) burrow test.

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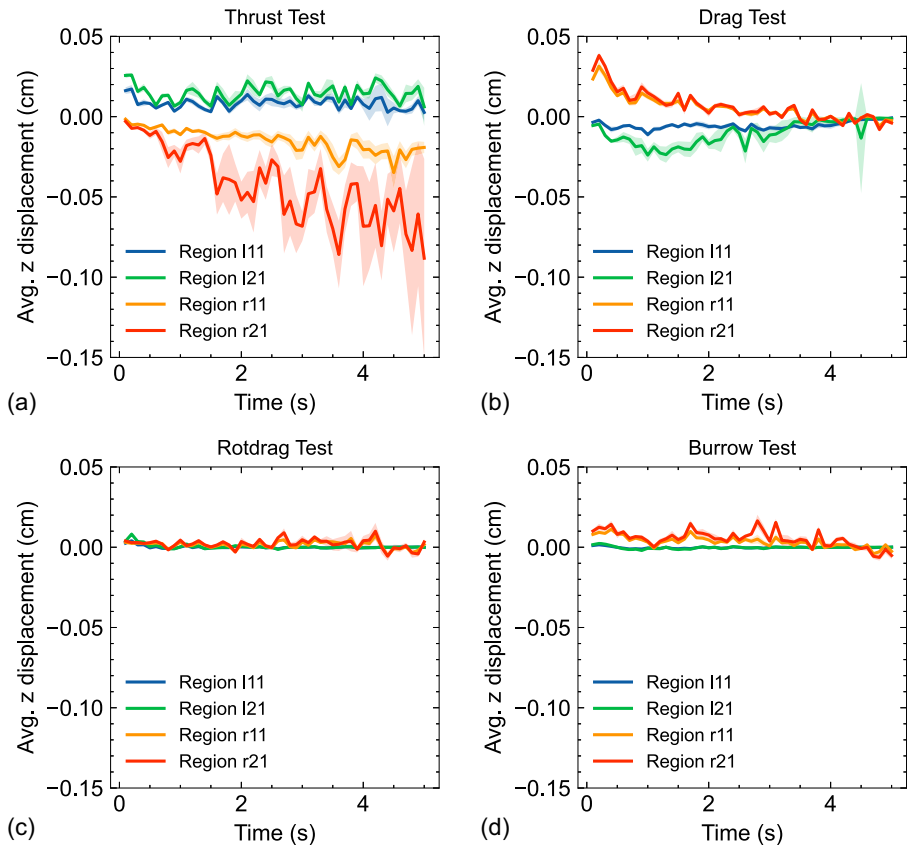

Fig. 15. Average vertical displacements of particles in four regions above the robot: (a) thrust test; (b) drag test; (c) rotdrag test; and (d) burrow test. The shaded areas indicate the standard deviations.

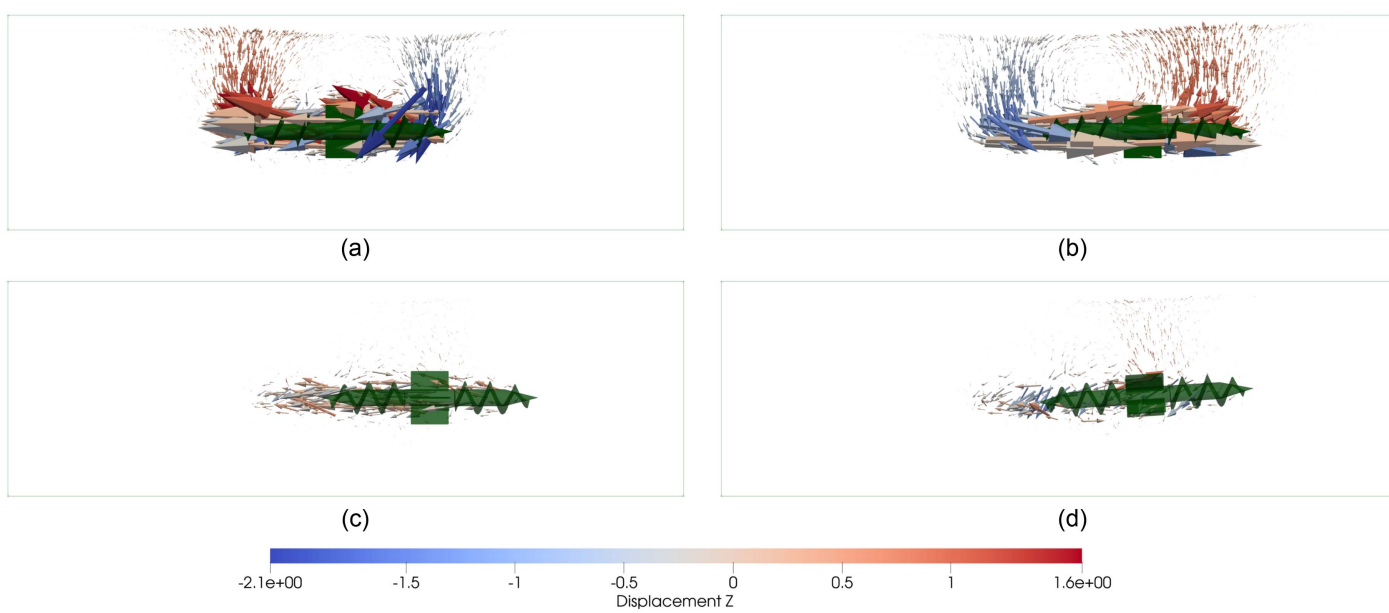

Fig. 16. Accumulated displacement vectors of particles around the robot (unit: cm): (a) thrust test; (b) drag test; (c) rotdrag test; and (d) burrow test.

in the upper left regions (l11, l21) experience upward movement. In both cases, particles closer to the robot (r21 and l21) exhibit more movement than those farther away. This observation aligns with the accumulated particle displacements [see Fig. 16(a)]: the upper right region supplies the auger with particles, which are then transported backward, pushing the existing particles in the upper left region further upward. The contact force network [Fig. 17(a)] shows that intensified forces are evident among particles in the wake regions of both augers, with particularly strong forces behind the back auger.

In the drag test [see Figs. 15(b) and 16(b)], the trends are almost opposite. As the robot moves, it ushes particles in its upper right region further upward. The forward movement of the robot creates

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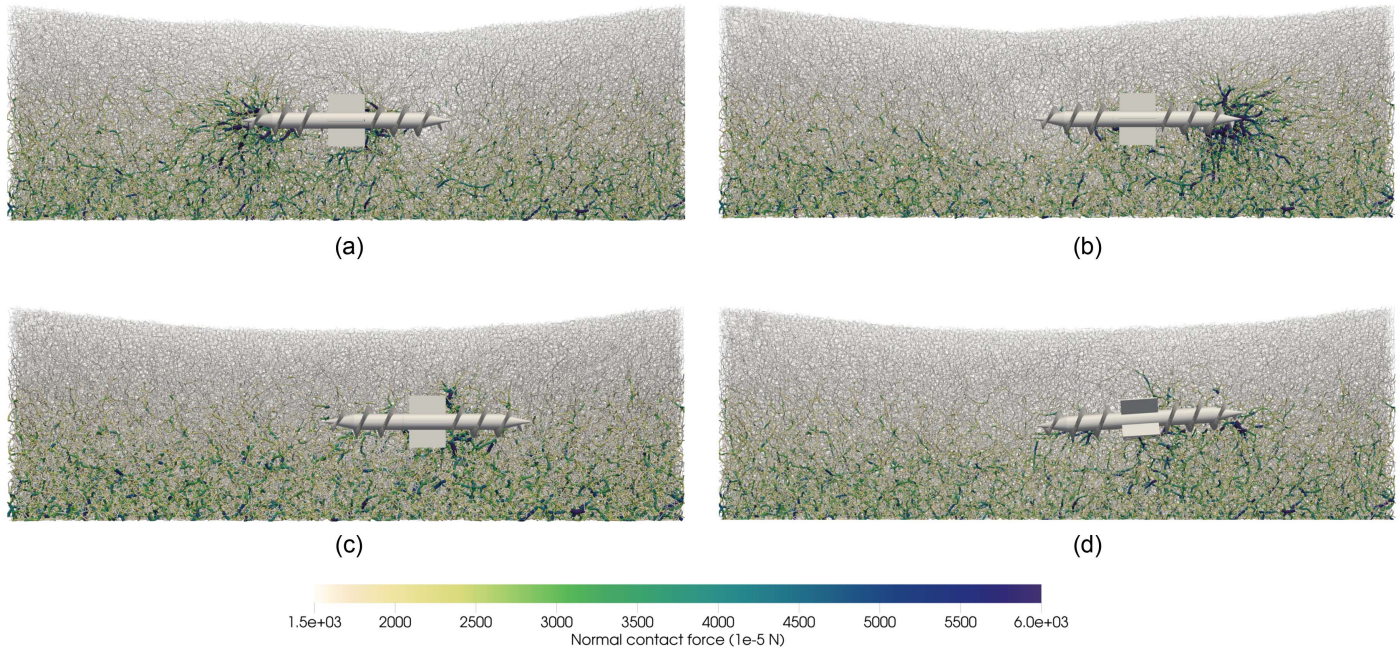

Fig. 17. Particle contact network: (a) thrust test; (b) drag test; (c) rotdrag test; and (d) burrow test.

a space at its left end, allowing particles in the upper left region to fill in. As the front auger passes through regions r11 and r22, the particles gradually shift back. This process repeats as the robot continues to move forward, dragging the particles immediately surrounding it along with it. The contact force network [Fig. 17(b)] shows that elevated forces mainly occur in front of the front auger.

In the rotdrag and burrow cases, the vertical movements of particles in all regions of interest are significantly smaller [see Figs. 15 (c and d)]. Only the particles directly surrounding (especially in front of) the robot are transported backward [see Figs. 16(c and d), and Supplemental Materials]. For the burrow case, a slight upward motion of particles in the upper right region of the robot is observed, which can be attributed to the slight pitching of the robot. The contact force networks for the rotdrag and burrow cases [Figs. 17(c and d)] are similar, though less pronounced than in the drag case.

The distribution of the contact force network and accumulated displacements (Figs. 16 and 17) indicates that there are no visible boundary effects, as particle forces and displacements near the container boundaries remain unaffected by the movement of the robot. This conclusion is further supported by the quantitative analysis of the influence zone provided in the Supplemental Materials.

Discussion

Thrust Redistribution and Drag Force Reduction

Comparing the forces across the kinematics-controlled cases and the burrow case suggests that the thrust and drag forces are influenced by the robot's movement.

The relative magnitudes of the forces on the left and right augers in all the cases as summarized in thrust and drag forces suggest that one or more of the following occurs when coupling auger rotation and translation:

1. The thrust force from the front right auger increases when the robot moves forward.
2. The thrust force from the left auger decreases when the robot moves forward.
3. The drag force on the front right auger decreases when the augers rotate.
4. The drag force on the left auger increases when the augers rotate.

Hypotheses 1 and 2 suggest thrust redistribution due to robot movement. The particle displacements shown in Fig. 16 indicate that the granular media in front of the robot transitions from an active state (moving toward the robot) in the Thrust case to a passive state (moving away from the robot) in the burrow case. This is also supported by the strong contact force regions shown in Fig. 17. The opposite behavior is observed for the granular media behind the robot. This shift provides a possible explanation for hypotheses 1 and 2: the thrust force increases on the front auger when the soil being transported shifts from an active state to a passive state, and it decreases on the back auger when the soil shifts from a passive state to an active state.

Hypotheses 3 and 4 suggest that auger rotation affects drag forces. It is well established that rotation typically reduces penetration resistance, which holds true for both smooth penetrators (Tang and Tao 2022a, b; Tang et al. 2024) and augers (Shaharear et al. 2023). Therefore, hypothesis 3 is likely valid, while hypothesis 4 lacks support. In contrast, auger rotation increases the drag forces on the stator (Fig. 12). In the drag test, the drag force on the stator is approximately -0.29 N, compared to about -0.98 N in the rotdrag test. In the thrust test, where the stator is fixed in place, a drag force of -0.56 N is recorded. This drag force results from the backward-moving particles transported by the front auger. This explains the increased drag force on the stator in the rotdrag test.

Lift Forces due to Drag and Axial Rotation

Lift forces in granular media arise from various mechanisms, each associated with specific types of motion:

1. Granular convection: Responsible for the Brazil nut effect (Knight et al. 1993), where larger particles in a granular mixture rise to the surface under vibration, defying gravity.

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2. Drag-induced lift: Observed when symmetrical objects are dragged through granular media (Ding et al. 2011; Maladen et al. 2011b). This phenomenon is also evident in the drag test of the current study.
3. Stirring-induced lift: Generated by rotating a rod in the plane of its long axis (Guillard et al. 2014).
4. Combined drag and rotation: Lift produced when a rod is simultaneously dragged and rotated along its long axis (Kumar et al. 2019), analogous to the magnus effect experienced by spinning balls moving through air.

While the latter two phenomena involve rotation, they also incorporate drag at the leading surfaces. These lift forces generally stem from changes in tangential force direction, uneven shearing of force chains, and asymmetric contact distribution across the object.

The current study reveals a novel type of lift force generated solely by rotation along a rod or auger's long axis, without any dragging motion. Preliminary results from ongoing experimental Thrust tests with an alternative auger design corroborate this finding, indicating that axial rotation of augers consistently produces a net upward lift force, regardless of rotation direction or auger handedness. This newly discovered lift mechanism warrants further, more systematic investigation to quantify and characterize its effect.

DEM versus DEM-MBD Simulations in Studying Burrowing

Simulating burrowing mechanisms accurately requires balancing computational efficiency with physical fidelity. Pure DEM has served as a valuable starting point for geotechnical researchers, offering computational speed and suitability for kinematics-controlled tests (e.g., thrust, drag, and rotdrag tests in this study). Researchers have further enhanced DEM for burrowing-related phenomena, incorporating force-limit servo control mechanisms for vertical burrowing (Chen et al. 2023, 2021; Zhang et al. 2023) and additional lift forces for horizontal penetration (Zhang and Fuentes 2022). These ad hoc advancements have provided important insights while highlighting areas for further refinement, such as torque balance and pitching dynamics. However, to comprehensively capture the physics of both granular media and moving structures, a coupled DEM-MBD cosimulation approach is preferred. This method effectively resolves granular-robot interaction forces and updates robot kinematics, overcoming limitations of pure DEM approaches.

A key challenge in burrowing research is predicting whether burrowing will occur given a specific robot design and kinematic control (e.g., rotation speed of augers). Based on the findings of this study, researchers should exercise caution when using a pure DEM approach for this purpose. It is generally inaccurate to assume that burrowing will occur simply because the total thrust force measured in a thrust test exceeds the total drag force in drag tests. While this argument might hold true for helical swimming, it fails to account for the redistribution of thrust and drag forces resulting from the coupled rotation and translation movement of the augers in burrowing scenarios. A more reliable indicator is the rotdrag test. If the total horizontal force on the robot in a rotdrag test is positive, it suggests that the robot would likely move forward in a burrowing test, and vice versa.

Limitations and Outlook

Despite its advantages, the adopted DEM-MBD framework (e.g., Chrono 7.0.0) lacks the capability to save individual particle-mesh contact forces for further analysis and may exhibit calculation instability during kinematics-controlled tests. In kinematics-controlled tests, a conflict between prescribed kinematics and the current Chrono MBD solver can introduce numerical variability. For such cases, we recommend to disable the MBD module and use DEM alone with imposed motion. These limitations highlight opportunities for future technical development.

In future studies, the DEM-MBD simulation framework can be extended to incorporate additional physics, such as deformable bodies. While the robotic structure in this study is assumed to be rigid, natural burrowers are often inherently soft or deformable. Similarly, many burrowing robot prototypes are designed using highly deformable materials (Das et al. 2023; Niiyama et al. 2022; Naclerio et al. 2021; Tao et al. 2020; Borela et al. 2021). In such cases, the DEM-MBD framework can be expanded by integrating a third module to account for the robot's flexibility and deformation. Continuous methods like the finite element method (FEM) are commonly used for this purpose, enabling comprehensive three-module cosimulations (Tasora et al. 2016).

Additionally, these cosimulation methods can be applied to systematically study the interactions between a dual-auger burrowing robot and granular media for optimization purposes. This includes investigating the effects of initial soil conditions such as relative density, particle size distribution, embedment depth, confining pressure, and anisotropy. Furthermore, they can be used to explore the influence of robot design and control parameters, including the geometric characteristics of the augers and stator, as well as the rotational speeds of individual augers.

Conclusion

This research employed a DEM-MBD cosimulation approach to model the burrowing behavior of a dual-auger robot. It was observed that when the augers rotated in the same direction, the robot primarily burrowed horizontally, with minor pitching and yawing motions. A stator was necessary to provide antitorque, causing it to roll in the opposite direction of the augers.

Complementary tests were conducted using both pure DEM and DEM-MBD cosimulation approaches. The control test, involving smooth rods without helical augers, confirmed that the robot's motion is attributable to the helical geometry of the augers. The kinematics-controlled tests, including thrust, drag, and rotary tests, offered insights into the thrust, drag, and lift forces responsible for the robot's burrowing motion. While pure DEM provides particle-contact modeling for kinematics-controlled scenarios, the DEM-MBD cosimulation provides a realistic, fully coupled description of burrowing mechanics by resolving both granular interactions and the robot's rigid-body dynamics.

It was concluded that while rotating augers generate thrust forces, these forces redistribute once the robot starts to move. Specifically, in a thrust test where the stator is fixed, the thrust force from the back auger is greater than that from the front auger, as it draws in nearby particles and transports them backward. However, when the robot moves forward, either through a prescribed rotdrag motion or induced burrowing, the thrust forces on the two augers likely redistribute, resulting in a decrease in thrust on the back auger and an increase on the front auger. It is also hypothesized that auger rotation reduces the drag forces experienced by the augers, as supported by related research. Burrowing becomes possible when thrust forces exceed drag forces. However, since both thrust and resistance are influenced by auger motion, the rotary Drag test serves as a more reliable indicator of burrowing potential. This test provides a better assessment of whether burrowing will occur for a given robot design and auger input motion, as it more accurately simulates the complex interactions between the robot and the surrounding medium. Additionally, the observed upward and pitching motions are attributed to

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the uneven distribution of lift forces induced by the dragging and rotating augers.

Appendix. MBD Formulations

Table 2 provides the material and interaction properties of the simulated particles and meshes. Based on the number of degrees of freedom for each joint, algebraic equations of motion can be applied to determine the behavior of kinematic joints as below, where $\mathbf{q}$ = vector of generalized coordinates, and t = time variable

$$\Phi(\mathbf{q}, t) = 0 \tag{7}$$

In addition, velocity and acceleration constraints can be derived by the first and second differentiation of the equation with respect to time (Marques et al. 2019; Kruggel-Emden et al. 2007)

$$\Phi_q \ddot{\mathbf{q}} = \boldsymbol{\gamma} \tag{8}$$

In which Φ_q is the Jacobian matrix of the constraint equations (A $3k$ by n matrix with k the number of contact points and n the degree of freedom), $\ddot{\mathbf{q}}$ is the vector of generalized accelerations, and $\boldsymbol{\gamma}$ represents the right-hand side of the equation which denotes the kinetic energy of joints configuration in terms of the square of velocity constraints (Bauchau et al. 2009). In addition, by applying Newton's second law of motion, which describes dynamic systems in terms of force and momentum, we have

$$M\ddot{\mathbf{q}} = \mathbf{g} \tag{9}$$

where M = global matrix of mass and inertia of the system; and $\mathbf{g}$ = resultant force of all the internal and external forces of the system (Marques et al. 2019). By applying the reaction forces of each joint, $\mathbf{g}^c$, to Lagrange's equation, we have

$$\mathbf{g}^c = (-\Phi_q)^T \boldsymbol{\Lambda} \tag{10}$$

where $\boldsymbol{\Lambda}$ = vector of the Lagrange multipliers for the maximum and minimum reaction force of each joint. Adding Eqs. (4) and (5), we can write the equation of motion of a generally constrained multibody as (Marques et al. 2019)

$$\begin{pmatrix} M & (-\Phi_q)^T \\ \Phi_q & 0 \end{pmatrix} \begin{pmatrix} \ddot{\mathbf{q}} \\ \boldsymbol{\Lambda} \end{pmatrix} = \begin{pmatrix} \mathbf{g} \\ \boldsymbol{\gamma} \end{pmatrix} \tag{11}$$

Table 2. Material properties of the simulated particles and meshes

Property	Particles	Mesh (robot) objects	Phase I	Phase II
	Material properties			
Particle radius (cm)	0.15	—	—	—
Density (g/cm^3)	2.55	1	—	—
Poisson's ratio	0.22	0.3	—	—
Young's modulus (Pa)	7×10^7	1.8×10^9	—	—
	Interaction properties			
Coefficient of restitution				
Sphere-sphere	—	—	0.22	0.22
Sphere-wall	—	—	0.22	0.22
Sphere-mesh	—	—	0.3	0.3
Coefficient of static friction				
Sphere-sphere	—	—	0.08	0.16
Sphere-wall	—	—	0.08	0.16
Sphere-mesh	—	—	0.15	0.3
Coefficient of rolling friction				
Sphere-sphere	—	—	0.0	2.5×10^{-5}
Sphere-wall	—	—	0.0	2.5×10^{-5}
Sphere-mesh	—	—	0.0	0.17

Data Availability Statement

Some or all data, models, or code generated or used during the study are available in a repository online in accordance with funder data retention policies.

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Author Contributions

Sarina Shahhosseini: Data curation; Formal analysis; Investigation; Methodology; Software; Validation; Visualization; Writing – original draft. Junliang "Julian" Tao: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Resources; Software; Supervision; Validation; Visualization; Writing – original draft; Writing – review and editing.

Notation

The following symbols are used in this paper:

$\mathbf{C_L C_R}$ = position vector from the centroid of the left auger to that of the right auger, respectively;
$\mathbf{C_S C_L}$, $\mathbf{C_S C_R}$ = position vectors from the stator centroid to the centroids of the left and right augers, respectively;
$[D_x, D_y, D_z]$ = displacement of the robot along x, y, and z axes;
$\mathbf{F}_L$, $\mathbf{F}_S$, $\mathbf{F}_R$ = forces acting on the left auger, stator, and right auger, respectively;
$\mathbf{F}_{total}$ = total force acting on the robot;
$[F_{x-total}, F_{y-total}, F_{z-total}]$ = components of the total force along the x, y, and z axes;
$\mathbf{G}_L$, $\mathbf{G}_S$, $\mathbf{G}_R$ = gravitational forces on the left auger, stator, and right auger, respectively;
$\mathbf{g}$ = resultant force of all internal and external forces of the system;
$\mathbf{g}^c$ = reaction forces at each joint;
M = global matrix of mass and inertia of the system;
$[P_{xL}, P_{yL}, P_{zL}]$ = left auger's centroid position at a given time along x, y, and z axes;
$[P_{xR}, P_{yR}, P_{zR}]$ = right auger's centroid position at a given time along x, y, and z axes;
$[P_{xS}, P_{yS}, P_{zS}]$ = Stator's centroid position at a given time along x, y, and z axes.
$[P_{xS0}, P_{yS0}, P_{zS0}]$ = Stator's original position along x, y, and z axes;
pitch = pitch angle of the robot, determined by the relative z-position of the left and right sides of the robot;
$\mathbf{q}$ = vector of generalized coordinates;
$\ddot{\mathbf{q}}$ = vector of generalized accelerations;
roll = roll angle of the robot, calculated by tracking the orientation of a vector perpendicular to the long axis of the robot;

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$\mathbf{T}_{total}$ = total torque acting on the robot;
$[T_{x-total}, T_{y-total}, T_{z-total}]$ = components of the total torque along the x, y, and z axes;
t = time variable;
$\mathbf{V}_0$, $\mathbf{V}_t$ = initial and current vectors used to calculate the roll of the robot;
yaw = yaw angle of the robot, determined by the relative x- and y-positions of the left and right sides of the robot;
$\boldsymbol{\gamma}$ = right-hand side of the acceleration equation, representing kinetic energy in terms of velocity constraints;
$\boldsymbol{\Lambda}$ = vector of Lagrange multipliers for the reaction forces;
$\Phi(\mathbf{q}, t)$ = equation of motion of kinematic joints; and
Φ_q = Jacobian matrix of the constraint equations.

Supplemental Materials

Supplemental material associated with this article are available online in the ASCE Library (www.ascelibrary.org).

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